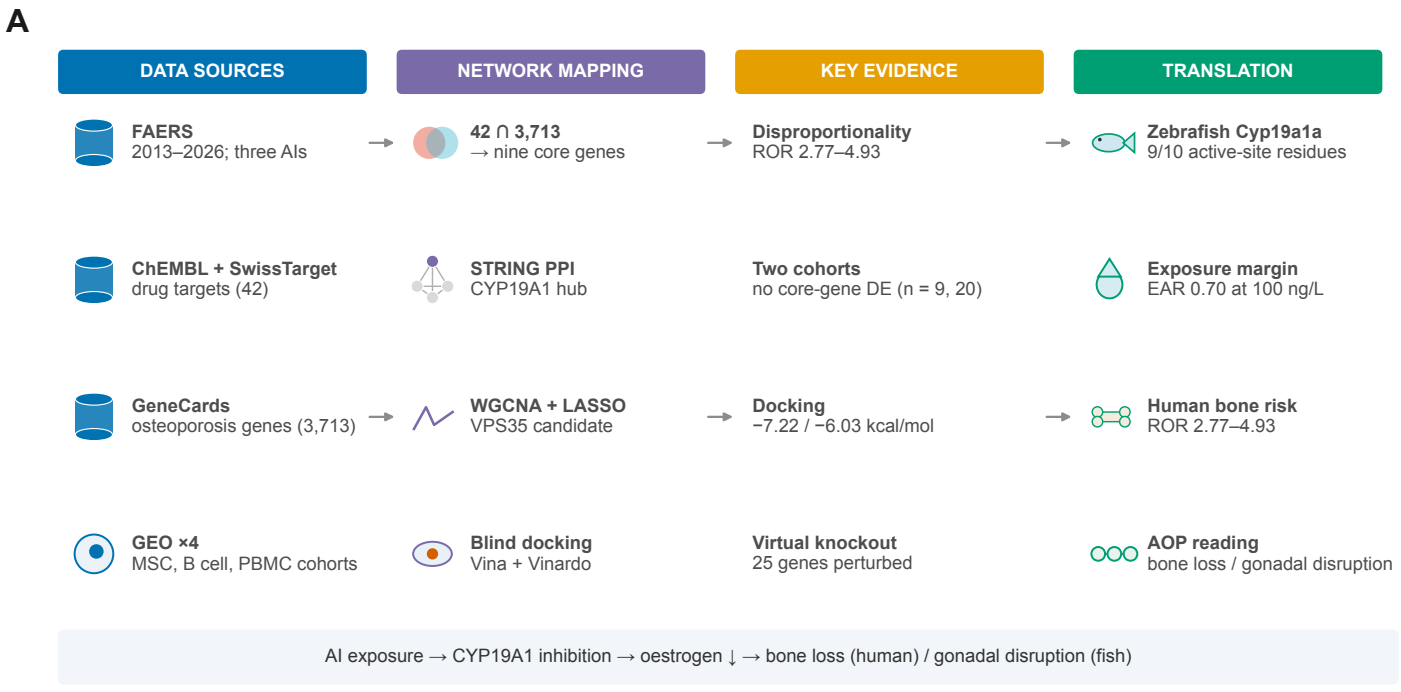
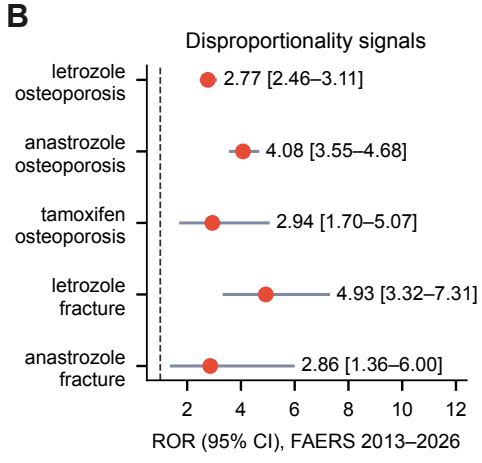
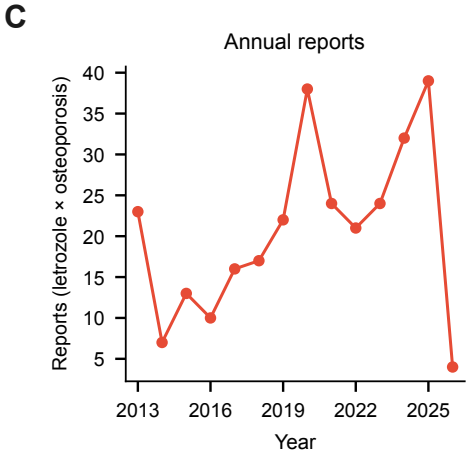
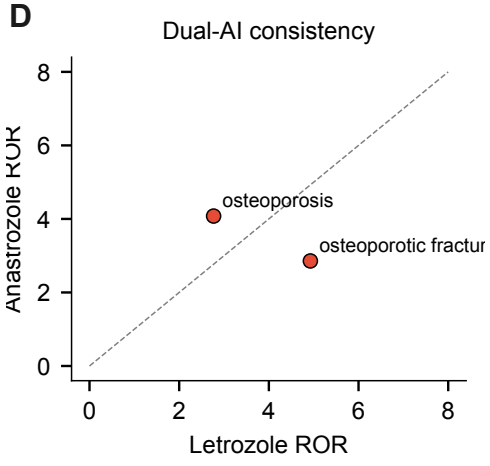
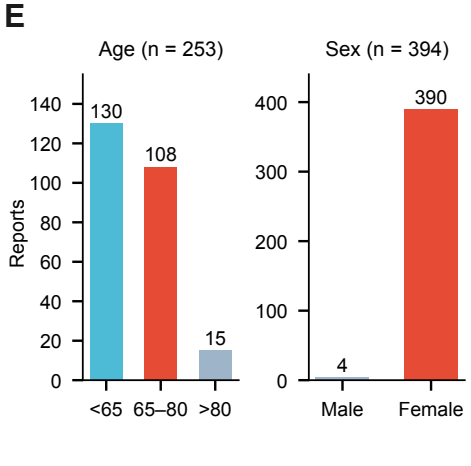
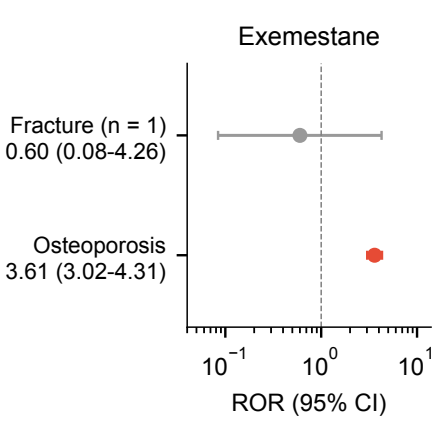

A
DATA SOURCES
NETWORK MAPPING
KEY EVIDENCE
TRANSLATION
FAERS
2013–2026; three AIs
42 ∩ 3,713
→ nine core genes
Disproportionality
ROR 2.77–4.93
Zebrafish Cyp19a1a
9/10 active-site residues
ChEMBL + SwissTarget
drug targets (42)
STRING PPI
CYP19A1 hub
Two cohorts
no core-gene DE (n = 9, 20)
Exposure margin
EAR 0.70 at 100 ng/L
GeneCards
osteoporosis genes (3,713)
WGCNA + LASSO
VPS35 candidate
Docking
−7.22 / −6.03 kcal/mol
Human bone risk
ROR 2.77–4.93
GEO ×4
MSC, B cell, PBMC cohorts
Blind docking
Vina + Vinardo
Virtual knockout
25 genes perturbed
AOP reading
bone loss / gonadal disruption
AI exposure → CYP19A1 inhibition → oestrogen ↓ → bone loss (human) / gonadal disruption (fish)


B
Disproportionality signals
letrozole osteoporosis
2.77 [2.46–3.11]
anastrozole osteoporosis
4.08 [3.55–4.68]
tamoxifen osteoporosis
2.94 [1.70–5.07]
letrozole fracture
4.93 [3.32–7.31]
anastrozole fracture
2.86 [1.36–6.00]
2
4
6
8
10
12
ROR (95% CI), FAERS 2013–2026


C
Annual reports
Reports (letrozole × osteoporosis)
40
35
30
25
20
15
10
5
2013
2016
2019
2022
2025
Year


D
Dual-AI consistency
Anastrozole ROR
8
6
4
2
0
osteoporosis
osteoporotic fractur
0
2
4
6
8
Letrozole ROR


E
Age (n = 253)
Sex (n = 394)
Reports
140
120
100
80
60
40
20
0
130
108
15
<65
65–80
>80
400
300
200
100
0
390
4
Male
Female


F
Exemestane
Fracture (n = 1)
0.60 (0.08-4.26)
Osteoporosis
3.61 (3.02-4.31)
10^−1
10^0
10^1
ROR (95% CI)